Technical Document #144: Data Engineering - Comprehensive Overview

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Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- Data warehouses store structured data optimized for analytical queries.

- Stream processing handles continuous data streams in real-time.

- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loa